

A p-stable solution of Casey’s metric-preserving dimension-reduction conjecture above a fixed scale

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1 Source and target

The source paper is Michael P. Casey, “Linear Dimension Reduction Approximately Preserving a Function of the 1-Norm”, arXiv:1906.03536, Electronic Journal of Statistics. On page 3, after proving the Cauchy case, the paper states the following conjecture for p-stable projections.

For $1 < p < 2$, replace the entries F_{ij} by iid standard p-stable random variables and set $\mu(\lambda) = \mathbb{E}\xi(\lambda F_{11})$. With F and μ modified as above, and ρ, ϵ, k as in Theorem 1.0.1, the bound

$$\mu\left(\frac{\|x - y\|_p}{1 + \epsilon}\right) \leq \rho(Fx, Fy) \leq \mu((1 + \epsilon)\|x - y\|_p)$$

holds for all $x, y \in P$ with $\|x - y\|_p = \Omega(1)$.

The paper defines

$$\rho(u, v) = \frac{1}{k} \sum_{i=1}^k \xi(|u_i - v_i|), \quad \xi(t) = \log(1 + \sqrt{t}) + \frac{1}{2} \log(1 + t), \quad t \geq 0.$$

Since ξ is defined on nonnegative arguments and the proof throughout uses $\xi(\lambda|W|)$, the statement below uses the consistent interpretation

$$\mu_p(\lambda) = \mathbb{E} \xi(\lambda|W|),$$

where W is a standard symmetric p-stable random variable.

for $\lambda > 0$. Our main theorem has several regimes depending on how big $\|x - y\|_1$ can be. (See theorems 3.0.1, 3.0.3, and 3.0.9.) However, the primary result is as follows.

Theorem 1.0.1. *Let F , ρ , and μ be as above. Given N points $P \subset \mathbb{R}^D$ and $\epsilon \in (0, 1)$,*

$$\mu\left(\frac{\|x - y\|_1}{1 + \epsilon}\right) \leq \rho(F(x), F(y)) \leq \mu((1 + \epsilon)\|x - y\|_1)$$

for all $x, y \in P$ with $\|x - y\|_1 \geq \sqrt{1 + \epsilon}$, provided

$$k = \frac{C}{\epsilon^2(1 - \epsilon)^2} \ln N.$$

Independent of its interest as an analog of the Johnson-Lindenstrauss lemma, theorem 1.0.1 also contributes to the study of p -stable projections. In fact, we make the following conjecture for $1 < p < 2$ upon replacing the entries F_{ij} of F by i.i.d. standard p -stable random variables and setting $\mu(\lambda) = \mathbb{E}\xi(\lambda F_{11})$. Just like the 1.0.1, the conjecture could have several parts based on how large $\|x - y\|_p$ is, but the primary conjecture is as follows.

Conjecture 1.0.2. With F and μ modified as above, and ρ , ϵ , and k as in theorem 1.0.1, the following bound holds

$$\mu\left(\frac{\|x - y\|_p}{1 + \epsilon}\right) \leq \rho(F(x), F(y)) \leq \mu((1 + \epsilon)\|x - y\|_p)$$

for all $x, y \in P$ with $\|x - y\|_p = \Omega(1)$.

The setup for the proof would be the same as for theorem 1.0.1, relying on 1st and 2nd moment estimates for $\xi(\lambda|W|)$; however, because the density for a p -stable random variable W is only implicitly defined, the needed 1st and 2nd

2 Result

Theorem 1 (Fixed-threshold p -stable form of the conjecture). *Fix $1 < p < 2$, $\lambda_0 > 0$, and $a > 0$. Let $P \subset \mathbb{R}^D$ be an N -point set, $N \geq 2$, and let $F : \mathbb{R}^D \rightarrow \mathbb{R}^k$ have iid standard symmetric p -stable entries, normalized so that $\mathbb{E}e^{itW} = e^{-|t|^p}$. Define ρ , ξ , and $\mu_p(\lambda) = \mathbb{E}\xi(\lambda|W|)$ as above.*

There is a constant $C = C(p, \lambda_0, a)$ such that, for every $0 < \epsilon < 1$, if

$$k \geq \frac{C}{\epsilon^2(1 - \epsilon)^2} \log N,$$

then with probability at least $1 - N^{-a}$, every pair $x, y \in P$ with $\|x - y\|_p \geq \lambda_0$ satisfies

$$\mu_p\left(\frac{\|x - y\|_p}{1 + \epsilon}\right) \leq \rho(Fx, Fy) \leq \mu_p((1 + \epsilon)\|x - y\|_p).$$

Thus the conjecture is proved under the standard interpretation of $\|x - y\|_p = \Omega(1)$ as a fixed positive lower bound. The proof does not claim uniformity when that hidden lower bound is allowed to tend to zero with N , ϵ , or p .

3 Proof intuition

The Cauchy proof in the source paper computes first and second moments of $\xi(\lambda|X|)$ explicitly. For general p -stable W , those explicit formulas are not available, but the special form of ξ gives a

simpler uniform route. Put

$$\phi(s) = \xi(e^s) = \log(1 + e^{s/2}) + \frac{1}{2} \log(1 + e^s).$$

Then $0 \leq \phi'(s) \leq 1$, so changing λ only translates a 1-Lipschitz function of $S = \log |W|$. The variable S has exponential tails in both directions: positive tails follow from finite q -moments of p -stable variables, and negative tails follow from boundedness of the stable density near zero. Therefore the centered variables $\xi(\lambda |W|) - \mu_p(\lambda)$ obey a Bernstein inequality uniformly in λ .

The remaining point is deterministic. Above a fixed scale λ_0 , the increase of μ_p from $\lambda/(1 + \varepsilon)$ to λ , and from λ to $(1 + \varepsilon)\lambda$, is at least a positive constant times ε . Uniform concentration within that gap for all pairs proves the conjectured inequalities.

4 Proof

Lemma 1 (Uniform sub-exponential concentration). *For every $1 < p < 2$ there are constants $K_p, c_p > 0$ such that, for all $\lambda > 0$, all $0 < t \leq 1$, and iid copies $W_1, \dots, W_k$ of W ,*

$$\mathbb{P} \left\{ \left| \frac{1}{k} \sum_{j=1}^k (\xi(\lambda |W_j|) - \mu_p(\lambda)) \right| > t \right\} \leq 2 \exp(-c_p k t^2).$$

Proof. Let $S = \log |W|$ and $\phi(s) = \xi(e^s)$. Since

$$\phi'(s) = \frac{1}{2} \frac{e^{s/2}}{1 + e^{s/2}} + \frac{1}{2} \frac{e^s}{1 + e^s},$$

ϕ is 1-Lipschitz. For $b = \log \lambda$,

$$|\phi(b + S) - \mathbb{E}\phi(b + S)| \leq |\phi(b + S) - \phi(b)| + \mathbb{E} |\phi(b + S) - \phi(b)| \leq |S| + \mathbb{E} |S|.$$

Choose any $0 < q < 1$. A standard symmetric p -stable variable has $\mathbb{E} |W|^q < \infty$, since $q < p$, and has a bounded continuous density near zero, hence $\mathbb{E} |W|^{-q} < \infty$. Consequently

$$\mathbb{E} e^{q|S|} = \mathbb{E} \max(|W|^q, |W|^{-q}) < \infty.$$

Thus the centered variables $\phi(b + S) - \mathbb{E}\phi(b + S)$ have sub-exponential norm bounded by a constant depending only on p , uniformly over $b \in \mathbb{R}$. Bernstein's inequality for averages of centered sub-exponential variables gives

$$\mathbb{P} \left\{ \left| \frac{1}{k} \sum_{j=1}^k (\xi(\lambda |W_j|) - \mu_p(\lambda)) \right| > t \right\} \leq 2 \exp\{-ck \min(t^2/K_p^2, t/K_p)\}.$$

Absorbing constants and using $0 < t \leq 1$ proves the lemma. $\square$

Lemma 2 (A fixed-scale gap for μ_p). *For every $1 < p < 2$ and $\lambda_0 > 0$, there is $\gamma = \gamma(p, \lambda_0) > 0$ such that, for all $\lambda \geq \lambda_0$ and $0 < \varepsilon < 1$,*

$$\mu_p((1 + \varepsilon)\lambda) - \mu_p(\lambda) \geq \gamma\varepsilon \quad \text{and} \quad \mu_p(\lambda) - \mu_p\left(\frac{\lambda}{1 + \varepsilon}\right) \geq \gamma\varepsilon.$$

Proof. For $t > 0$, set

$$h(t) = t\xi'(t) = \frac{1}{2} \frac{\sqrt{t}}{1 + \sqrt{t}} + \frac{1}{2} \frac{t}{1 + t}.$$

The function h is increasing, bounded, and positive on $(0, \infty)$. For $r > 0$ and $A > B > 0$,

$$\xi(Ar) - \xi(Br) = \int_B^A h(ur) \frac{du}{u}.$$

By boundedness of h , Fubini applies. With $A = 1 + \varepsilon$,

$$\mu_p(A\lambda) - \mu_p(\lambda) = \int_1^A \mathbb{E}h(u\lambda | W|) \frac{du}{u} \geq \log A \mathbb{E}h(\lambda_0 | W| / 2),$$

where the last lower bound is deliberately weakened for later symmetry. Likewise,

$$\mu_p(\lambda) - \mu_p(\lambda/A) = \int_{1/A}^1 \mathbb{E}h(u\lambda | W|) \frac{du}{u} \geq \log A \mathbb{E}h(\lambda_0 | W| / 2),$$

because $A < 2$, $u \geq 1/A$, and $\lambda \geq \lambda_0$. Since $\log(1 + \varepsilon) \geq \varepsilon/2$ for $0 < \varepsilon < 1$, the result follows with

$$\gamma = \frac{1}{2} \mathbb{E}h(\lambda_0 | W| / 2) > 0.$$

□

Proof of Theorem 1. Fix a pair $x, y \in P$, put $v = x - y$, and let $\lambda = \|v\|_p$. By p -stability, each coordinate of Fv has the same law as λW , independently over the k coordinates. Hence

$$\rho(Fx, Fy) \stackrel{d}{=} \frac{1}{k} \sum_{j=1}^k \xi(\lambda | W_j|).$$

Assume $\lambda \geq \lambda_0$. Lemma 2 gives a gap $\gamma\varepsilon$ from $\mu_p(\lambda)$ to each of the two desired boundary values. Therefore the event

$$|\rho(Fx, Fy) - \mu_p(\lambda)| \leq \gamma\varepsilon$$

implies

$$\mu_p\left(\frac{\lambda}{1 + \varepsilon}\right) \leq \rho(Fx, Fy) \leq \mu_p((1 + \varepsilon)\lambda).$$

Since $0 < \gamma\varepsilon \leq 1$, after reducing the concentration constant if necessary, Lemma 1 gives for this fixed pair

$$\mathbb{P}\{\text{the desired inequality fails for } x, y\} \leq 2 \exp(-c_p \gamma^2 k \varepsilon^2).$$

There are fewer than $N^2/2$ unordered pairs. Choosing

$$k \geq C(p, \lambda_0, a) \varepsilon^{-2} \log N$$

with C large enough makes the union bound at most N^{-a} . Since $(1 - \varepsilon)^{-2} \geq 1$, the displayed dimension bound in the theorem is a slightly weaker sufficient condition with the same dependence as Casey's Theorem 1.0.1. This proves the simultaneous assertion. □

5 Verification and novelty notes

The proof uses only standard p-stable facts: the scaling identity $\sum_j v_j W_j \stackrel{d}{=} \|v\|_p W$, finite moments $\mathbb{E}|W|^q < \infty$ for $q < p$, and boundedness of the p-stable density near zero. These are the same stable-distribution background facts invoked by the source paper and by standard references on p-stable projections.

Bounded novelty search performed before promotion:

- local run indexes were searched for arXiv:1906.03536, the exact title, “p-stable”, “metric preserving”, and “Conjecture 1.0.2”;
- the parsed source TeX and bibliography for arXiv:1906.03536 were inspected;
- web/arXiv searches were run for exact-title and close-variant phrases, including “p-stable metric preserving dimension reduction”, “Casey p-stable conjecture”, and “Cauchy projections metric preserving function”.

No exact later answer or duplicate packet was found in these bounded searches.

Human-review recommendation: review as a likely valid full solution to the fixed-threshold reading of Conjecture 1.0.2. The main points to check are the uniform sub-exponential domination by $|\log |W||$, the gap estimate in Lemma 2, and the interpretation of $\Omega(1)$.

References

- [1] Michael P. Casey. Linear dimension reduction approximately preserving a function of the 1-norm. *Electronic Journal of Statistics*; arXiv:1906.03536.
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